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Approximate analysis of roll force in a round–oval–round pass rolling sequence

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Abstract

This paper presents a semi-analytical method for the prediction of roll force in the oval–round (or round–oval) pass rolling sequence most commonly employed in present continuous rod (or bar) mills. Introduced here is the concept of a weak plane-strain deformation taking into account the deformation characteristics of the workpiece (material) in the roll grooves when the equation for the roll force is formulated. The flow stress (deformation resistance) of the workpiece is characterized by Shida's constitutive equation. Hot rod rolling experiments were conducted to examine the proposed semi-analytical method. The results reveal that using the weak plane-strain deformation condition is appropriate in solving three-dimensional problems, where it is often necessary to reduce the cost of computation by simplification. For low plain-carbon steel, the roll forces predicted are in good agreement with those obtained experimentally with the rolling temperature changed (800–1100°C). © 2001 Elsevier Science B.V. All rights reserved.

Keywords: Rod rolling; Roll force; Semi-analytical method; Weak plane-strain condition

1. Introduction

To predict the roll (separating) force at each pass (stand), a number of numerical studies have been presented on the basis of three-dimensional finite element techniques [1,2]. The finite element method is very effective in calculating the distribution of plastic deformation of a workpiece and the roll force, but considering the computational time and the complicated mechanical/thermal boundary conditions (the friction condition at the roll/material interface and the heat-transfer coefficients, which depend on the temperature and roll pressure), development of an analytical method is also desired for practical usage.

Obtaining a full analytical solution for the roll force has been one of the most difficult problems in the analysis of groove rolling because of the difficulties in formulating the governing differential equation depicting the complicated three-dimensional deformation behavior of the workpiece. Thus, there have been attempts to develop a simplified analytical method as an alternative.

In the simplified analytical method, the equation for the roll force together with the formulae for the strain, the strain rate, the average contact stress and the projected contact area in plate rolling theory were used with fitting constants on the basis of wide-ranged experimental data. These approaches, mostly based on the empirical modifications of plate rolling theory, have been reviewed [3,4].

Shinokura and Takai [5] presented an experimentally based model for calculating the roll force and torque using a simple compression test and non-dimensional parameters represented by the ratio of the projected contact area to the initial and final cross-sectional area of the workpiece deformed at a given pass. The non-dimensional parameters were obtained by regression of experimental data after extensive hot rod-rolling experiment.

In this paper, is presented a semi-analytical model to determine the roll force in the oval–round (or round–oval) pass rolling sequence most commonly employed in rod (or bar) mills. The semi-analytical model excludes any needs to conduct a preliminary hot rod rolling experiment to determine a coefficient. The idea of the proposed model is to reduce a three-dimensional problem into a two-dimensional one by introducing the concept of a *weak* plane-strain deformation condition which takes into account the deformation characteristics of the workpiece during groove rolling. The primary objective of introducing this concept is to treat a three-dimensional deformation problem effectively.

The background for this concept is that, during rolling, the deformation of the workpiece in the lateral (roll axis) direction is constrained due to the roll groove shape. Consequently, the deformation condition of the workpiece at the

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inside of the roll groove can be considered as a semi-plane-strain deformation condition in a way, even though the ratio of thickness to width of the cross-section of the workpiece is not small enough. Then, the equation for average contact stress (roll pressure) in groove rolling can be formulated by introducing a linear compensating function into that for plate rolling. This might seem to be an over-simplification but the result becomes very fruitful in solving the three-dimensional deformation problem.

The validity of the model has been examined by a round–oval–round pass hot rolling experiment with the temperature of the workpiece being changed. A two-high single stand pilot mill was used to remove any possibility of front (or back) tension on the workpiece.

In the following, the formulation procedure for the calculation of the roll force is described. It consists of two stages. The first stage is to obtain the deformation parameters. If the temperature of the workpiece is known at a given pass, the deformation parameters might be: (i) the cross-sectional shape of the workpiece at a given pass; (ii) the strain defined as the maximum average effective (equivalent) plastic strain at a given pass; (iii) the strain rate defined as the strain divided by the time required for the workpiece to undergo this strain in the roll gap. It should be noted that the deformation parameters are pre-requisites for the calculation of the roll force, which is the second stage. This includes: (i) the projected contact area; (ii) the average contact stress (roll pressure).

2. Deformation parameters

2.1. Prediction of cross-sectional shape

The first step in the analysis of rod rolling is predicting the cross-sectional shape of the workpiece deformed at a pass. The cross-sectional shape of an outgoing workpiece is predicted by the surface profile prediction model which is based on a weighting function and the linear interpolation of the radius of the curvature of an incoming workpiece and that of the roll groove to the roll axis direction. The formulation procedure of the surface profile prediction model and its experimental validation are described in [6].

Fig. 1 shows the rolling conditions (dimension of grooves and gaps) when the experiment was carried out. Fig. 2 describes the predicted surface profiles of the deformed workpiece and the measured values. C_x and C_y represent the co-ordinates where the roll groove and the deformed workpiece are separated.

2.2. Mean effective strain

Since the cross-sectional shape of the workpiece during rolling is not rectilinear, one encounters difficulties in formulating the equation for the mean effective strain at a pass. A way to overcome the difficulties may be by employing the equivalent rectangle approximation method that transforms a curved cross-section into a rectilinear one while the net cross-sectional area is maintained (Fig. 3). Section A–A, B–B and C–C correspond to the position where a section of the workpiece is about to be rolled, deformed and leaves the roll groove, respectively. Then, the mean effective strain

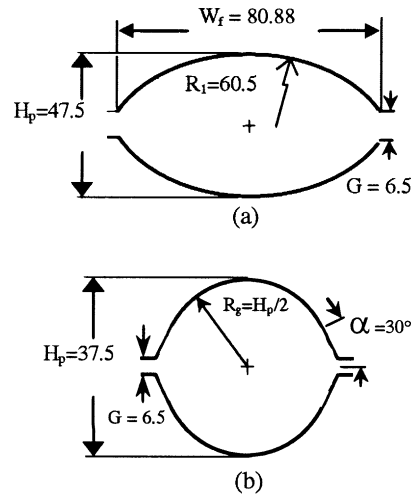


Fig. 1. Oval–round pass sequence employed in hot rod rolling experiments (mm): (a) oval pass; (b) round pass.

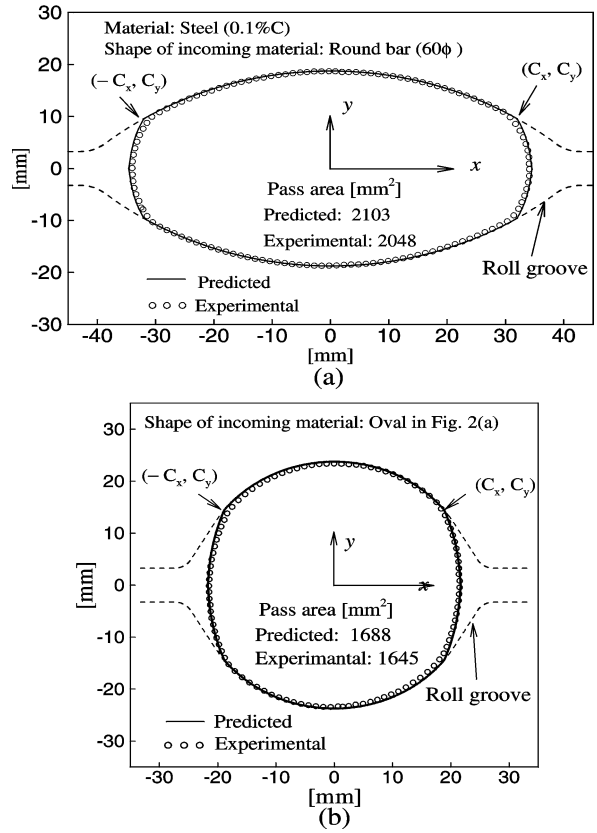


Fig. 2. Predicted and measured surface profiles when a specimen with 60 mm diameter is rolled into: (a) oval pass; (b) round pass as described in Fig. 1.

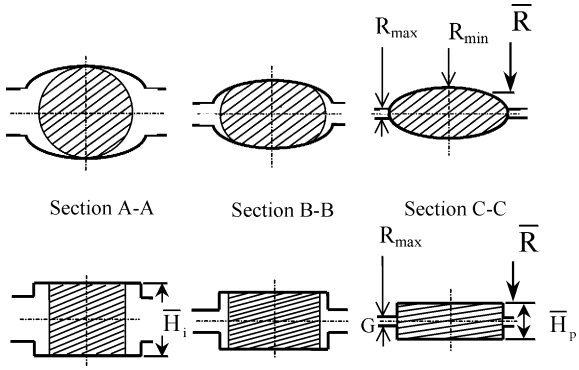


Fig. 3. Schematic description of equivalent rectangle approximation to round–oval pass rolling and effective roll radius $\bar{R}$, and equivalent cross-sectional heights, $\bar{H}_i$ and $\bar{H}_p$.

at a given pass can be calculated from the rectilinear shape of the workpiece transformed.

The equivalent rectangle approximation described in Fig. 3 can also be applied to oval–round pass rolling. With the hypothesis of parallelepiped deformation of the material, the incremental plastic deformation along each principal axis can be assumed to be proportional [7], i.e.

$$d\varepsilon_1 : d\varepsilon_2 : d\varepsilon_3 = \varepsilon_1 : \varepsilon_2 : \varepsilon_3. \tag{1}$$

From Fig. 4, the principal plastic strain for each axis can be defined as follows:

$$\text{Principal plastic strain along the } x\text{-axis}: \quad \varepsilon_1 = \ln\left(\frac{\bar{W}_i}{\bar{W}_p}\right), \tag{2}$$

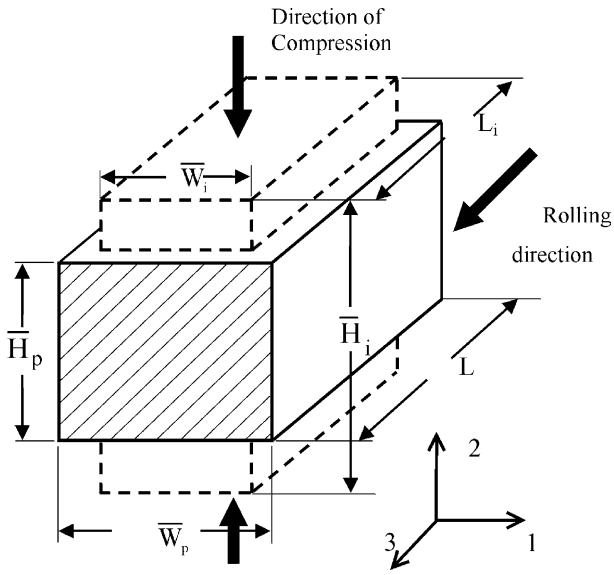


Fig. 4. Schematic representation of parallelepiped deformation of the equivalent rectangular section, $\bar{W}_i \times \bar{H}_i$, for an oval section (before the pass) and the equivalent rectangular section, $\bar{W}_p \times \bar{H}_p$, for a round section (after the pass).

$$\text{Principal plastic strain along the } y\text{-axis}: \quad \varepsilon_2 = \ln\left(\frac{\bar{H}_i}{\bar{H}_p}\right), \tag{3}$$

$$\text{Principal plastic strain along the } z\text{-axis}: \quad \varepsilon_3 = \ln\left(\frac{L_i}{L_p}\right). \tag{4}$$

The positive sign is defined as when the workpiece contracts. Thus, the sign of ε_2 is always positive due to the contraction along the y-axis, while the sign of ε_1 and ε_3 is negative due to the extension along the x-axis and the z-axis. Note that Eqs. (2)–(4) are valid under the condition that the principal plastic strains in the three principal axes are independent of each other. From the volume constancy condition, we have

$$\varepsilon_3 = -\varepsilon_1 - \varepsilon_2. \tag{5}$$

Then, the mean effective strain at a given pass, $\bar{\varepsilon}_p$, is:

$$\bar{\varepsilon}_p = \frac{2}{\sqrt{3}}\varepsilon_2\left[1 + \left(\frac{\varepsilon_1}{\varepsilon_2}\right)^2 + \left(\frac{\varepsilon_1}{\varepsilon_2}\right)\right]^{1/2}. \tag{6}$$

Here Eq. (6) is a closed-form solution for the mean effective strain in rod rolling, based on the parallelepiped deformation model. It shows that the mean effective strain in rod (or bar) rolling becomes equivalent to that in plate rolling if the ratio $(\varepsilon_1/\varepsilon_2)$ is negligibly small.

2.3. Mean effective strain rate

Once the mean effective strain is obtained, the mean effective strain rate is computed easily. The mean effective strain rate in rod rolling can be defined as the mean effective strain over a time interval which, on average, can be calculated from

$$\dot{\bar{\varepsilon}}_p = \frac{\bar{\varepsilon}_p}{t_p}, \tag{7}$$

where t_p represents the time interval taken for a section A–A to pass up to a section C–C in Fig. 3. The time interval can be expressed as

$$t_p = \frac{2\pi N\bar{R}}{60\bar{L}}\ (\text{s}), \quad \frac{60\bar{L}}{2\pi N\bar{R}} \tag{8}$$

where $\bar{R}$, $\bar{L}$ and N are, respectively, the effective roll radius, the effective projected contact length of the grooved roll and workpiece, and the rolling rpm at a given pass. The effective roll radius is calculated by the method of maximum width [3] when the cross-sectional area of the workpiece at a pass is obtained. The effective projected contact length can be expressed as:

$$\bar{L} = \sqrt{\left\{R_{max} - \tfrac{1}{2}(\bar{H}_p - G)\right\}(\bar{H}_i - \bar{H}_p)}. \tag{9}$$

3. Formulae of roll force

In the following, is described the formulation procedure for the computation of the roll force which includes the modeling of the: (i) projected contact area; (ii) average contact stress (roll pressure). Once these are acquired, the roll force at a given pass, F, is calculated by means of Eq. (10):

$$F = PA_{\mathrm{d}}, \tag{10}$$

where P represents the average contact stress applied to the workpiece and A_{d} the projected contact area.

3.1. Projected contact area

Modeling of projected contact area is an essential part of the prediction of roll force. To be consistent with the analytical method in the formulation of the deformation parameters, the equation for the projected contact area is also formulated by using a function that can be integrated analytically.

Fig. 5 shows the shape of the workpiece in contact with the work rolls and its geometrical designation. These were acquired by the emergency stop of a hot rod rolling pilot mill. The white line distinguishes the portion in contact.

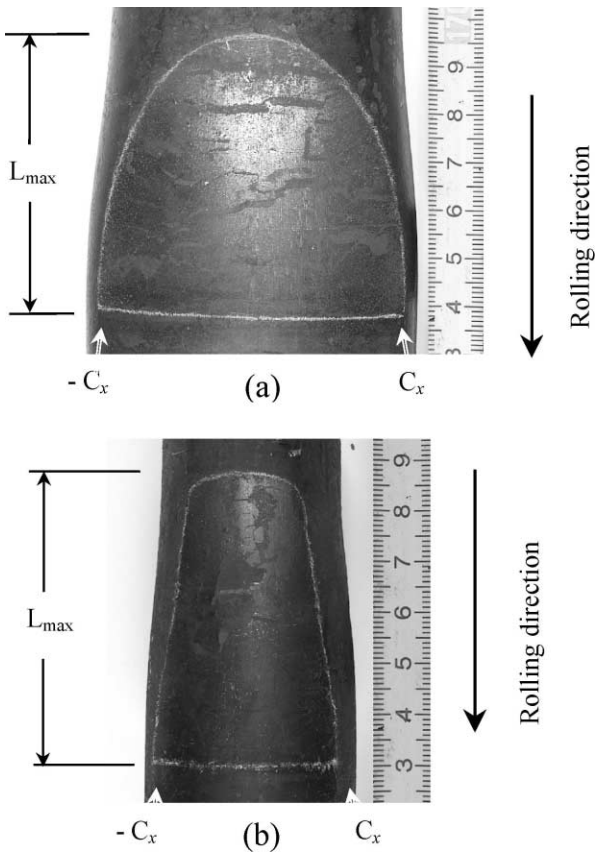


Fig. 5. Shape of the projected contact area of the workpiece deformed at the inside of the roll groove and its geometric designation (mm): (a) round–oval pass; (b) oval–round pass.

Table 1
Comparison of formulae for projected contact area

Present work	Shinokura and Takai [5]
$\frac{3}{2}L_{\max}C_x$	$\frac{\pi}{2}L_{\max}C_x$

Then, the projected contact area, A_{d}, is expressed as

$$A_{\mathrm{d}} = 2\int_0^{C_x} L_{\max}\left(1-\frac{x}{C_x}\right)^m \mathrm{d}x, \tag{11}$$

where m is a dimensionless shape factor. $m = \frac{1}{3}$ was chosen in this study. Here, C_x and $-C_x$, are the abscissa where the roll groove and deformed workpiece are separated (see Fig. 2). $L_{\max}$ represents the maximum distance of the projected contact area in the rolling direction.

For the purpose of comparison, the projected contact area modeled by Shinokura and Takai [5] is shown in Eq. (12):

$$A_{\mathrm{d}} = 2\int_0^{C_x} L_{\max}\left(1-\frac{x^2}{C_x^2}\right)^{1/2} \mathrm{d}x. \tag{12}$$

Table 1 compares the closed-form solution of the projected contact area modeled by Shinokura and Takai [5] with that modeled in the present work.

Table 2 demonstrates that the projected contact area predicted by the model proposed in the present work slightly underestimates the measured value. At the same time, the projected contact area predicted by Shinokura and Takai's model overvalues the measured value. It is worth noting that even though the shape of the projected contact area is different for each pass (Fig. 5), a single formula is enough to calculate the projected contact area for both passes.

3.2. Average contact stress (roll pressure)

In general, the workpiece between the work rolls in plate rolling has a three-dimensional deformation behavior, but the plane-strain condition has been applied successfully. This condition has been used in practice when the ratio of height to width of the cross-section of the workpiece is small enough such that the principal strain in the roll axis (gap) direction is negligible. Based on this assumption, the equation for the average contact stress under plane-strain deformation conditions is expressed as [8]:

$$P_{\mathrm{plate}} = K_{\mathrm{m}}\exp\left[\frac{\mu L}{h_{\mathrm{m}}}-1\right] \approx K_{\mathrm{m}}\exp\left[\frac{\mu L}{2h_{\mathrm{m}}}\right], \tag{13}$$

where μ is the Coulomb friction coefficient, K_{m} the average deformation resistance of the workpiece, L the projected

Table 2
Comparison of the predicted projected contact area and that measured

Pass sequence	Measured (mm^2)	Present work (mm^2)	Shinokura and Takai [5] (mm^2)
Round–oval	3036	2843	3507
Oval–round	1651	1633	1710

contact length between the flat roll and the workpiece, and h_m the mean height of the workpiece.

Here, a new term necessary for the calculation of average contact stress in rod (or bar) rolling is defined. Suppose that the deformation condition of the workpiece in plate rolling is defined as a *strong* plane-strain deformation condition. Then, that in rod rolling might be defined as a weak plane-strain deformation condition. The background for this definition is that, as illustrated in Fig. 3, the deformation of the workpiece in the lateral (roll gap) direction is constrained due to the groove shape. As a result, in a way the deformation condition at the inside of the roll groove can be considered as a semi-plane-strain deformation condition. In other words, the deformation condition of the workpiece at section C (Fig. 3) is not under a fully plane-strain condition, but might be under a weak plane-strain deformation condition.

Referring to Fig. 3 and Eq. (6), the description of the new definition can be summarized as follows:

Strong plane-strain deformation condition: $\varepsilon_1 = 0$.
Weak plane-strain deformation condition: $\varepsilon \neq 0$.

This definition will be useful in reducing a three-dimensional deformation problem to two-dimensional one. At this point, it is known that the average contact stress (roll pressure) in rod rolling ($\varepsilon_1 \neq 0$) is always smaller than that in plate rolling ($\varepsilon_1 = 0$) because the non-zero state of principal strain in the roll axis direction, i.e., $\varepsilon_1 \neq 0$ makes the deformation of the workpiece much easier than the zero state of principal strain in the roll axis direction, i.e., $\varepsilon_1 = 0$. Thus, with the weak plane-strain deformation condition, the average contact stress in an oval–round (or round–oval) pass rolling sequence can be formulated by modifying Eq. (13) as

$$P_{\text{rod}} = (1 - \varepsilon_1)\, K_m \exp\left[\frac{\mu \bar{L}}{\bar{h}_m}\right], \tag{14}$$

where $\bar{L} = [\{R_{\max} - \frac{1}{2}(\bar{H}_p - G)\}(\bar{H}_i - \bar{H}_p)]^{1/2}$ is the effective projected contact length between the grooved roll and the workpiece, and $\bar{h}_m = \left[\frac{1}{2}(\bar{H}_i + \bar{H}_p)\right]$ the effective mean height of the workpiece. The term $(1 - \varepsilon_1)$ in Eq. (14) plays a role as a linear compensating function, taking into account the change of deformation condition. The friction coefficient was set as 0.35.

3.3. Deformation resistance (flow stress) of the workpiece

Numerous constitutive equations, which predict the deformation resistance behavior of steels at elevated temperatures, have been presented and are reviewed elsewhere [9]. The constitutive equation should include all the possible rolling parameters and the material parameters for an accurate prediction of the deformation resistance.

In this study, the flow stress of the workpiece was characterized by Shida's constitutive equation [10], giving the flow stress as a function of the carbon content, the strain, the strain rate and the temperature. Shida's equation was chosen primarily due to its simplicity in application. This implies that since Shida's constitutive equation provides the flow stress of plain carbon steel the carbon contents and the rolling variables (strain, strain rate and temperature) of the material should be determined.

The equation also takes into account the deformation behavior of the steels in the austenite, ferritic and two-phase regions. The range of validity of the equation is quite broad. This equation is applicable for plain carbon steel in the range of carbon content 0.07–1.2%, temperature 700–1200°C, strain rate 0.1–100 s^{-1} and strain up to 0.7.

4. Experiment

4.1. Rolling equipment and specimen preparation

A two-high laboratory mill with a single stand was employed, driven by a 75 kW constant torque, DC motor. A ductile casting iron (DCI) roll was used, with 310 mm maximum diameter and 320 mm face width. The rolling speed was set at 34 rpm. A box type furnace with a working temperature of 1400°C was employed to heat the specimens to the desired rolling temperature.

Plain carbon steel (0.1%C) was obtained in the form of hot-rolled squares of 160 mm sides. The specimens to be rolled were cut and machined into round bar of 60 mm diameter and 300 mm length.

4.2. Experimental procedure

For the hot rod rolling experiments, the specimens were soaked at 1130°C for 1 h to ensure a homogeneous temperature distribution. They were then taken out of the furnace, held until the desired temperature was reached and then rolled. When the center temperature of the specimens reached 1100, 950 and 800°C, the tests were conducted. To measure the rolling temperature of the workpiece, a thermocouple (type K) with 1.6 mm outside diameter was embedded in 50 mm deep holes drilled in the tail ends of the specimen. A specimen was first rolled into the oval pass (Fig. 1(a)), at the desired temperature. The workpiece rolled at the oval pass was re-heated to the desired temperature before being rolled in the next pass. The workpiece was rolled into the round pass (Fig. 1(b)), with a rotation of 90° along the length direction. The roll forces were measured and using two force transducers during rolling, located over the bearing blocks of the top work roll.

5. Results and discussion

5.1. Time-history of the temperature distribution

The temperature of the specimen is uniform whilst in the closed furnace. As soon as it is removed from the furnace,

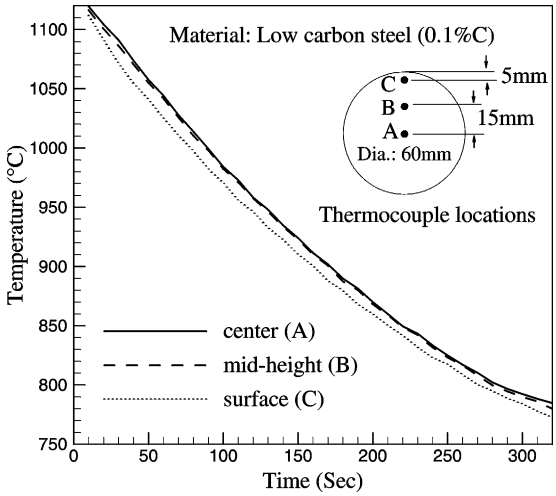


Fig. 6. Temperature history of the specimen cooled in air after it had been removed from the furnace.

the surface begins to cool at a faster rate than that at the center. The properties of the specimens depend on the cooling rate in addition to the heated temperature and since the former varies across the cross-section, homogeneous properties will not be obtained. Therefore, the average temperature of the specimen should be used when the deformation resistance of the specimen is needed. To measure the average temperature of the specimen, three thermocouples were embedded in the specimen, at the locations shown in Fig. 6. One of the thermocouples is at the center of the specimen, marked (A), another (B) at the mid-height and the third (C) placed as close as to the surface as possible. A typical temperature history during air cooling is shown in Fig. 6. The cooling rates at two locations, i.e., mid-height and at the center show almost no difference.

5.2. Comparison of measured and predicted roll force

Fig. 7 illustrates the comparison of measured and predicted roll forces for the oval pass and round pass, respectively. The roll forces predicted by the semi-analytical method developed in this study overestimated slightly those measured experimentally, the differences at each temperature being in the range 0.8–5.7%, on average. Considering the weak plane-strain assumption introduced for the calculation of roll force, this result is reasonable. In the round pass, some fluctuations in the measured roll force are observed. This might be due to an unstable rolling of the workpiece with an oval shape in the round pass, compared with the relatively stable rolling in the oval pass. Overall, the roll forces predicted by the proposed method are in good agreement with those measured. This demonstrates that the proposed semi-analytical method has simplicity and accuracy for practical usage.

On the other hand, as observed, Shinokura and Takai's method [5] is not capable of accurately predicting the roll

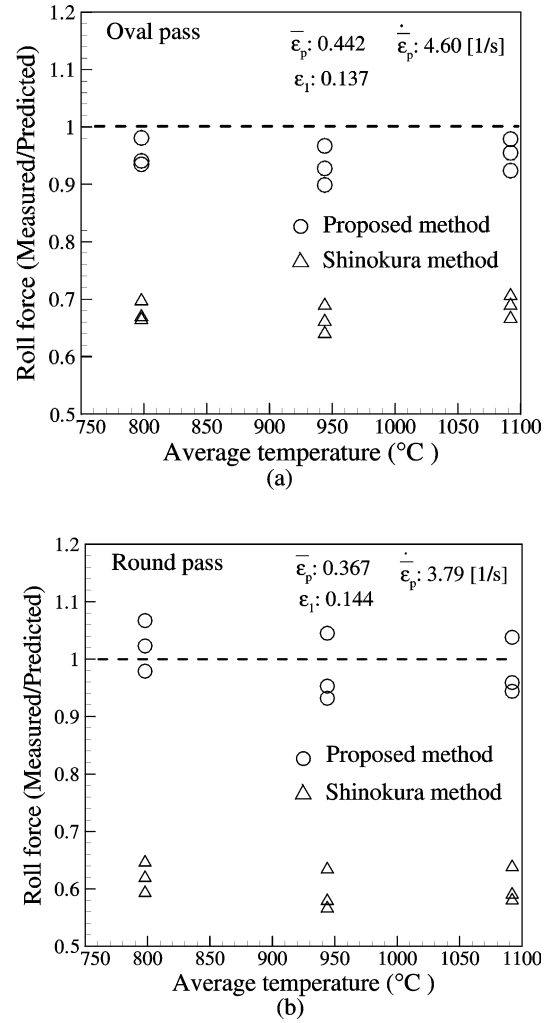


Fig. 7. Comparison of the predicted and measured roll forces for: (a) an oval pass; (b) a round pass.

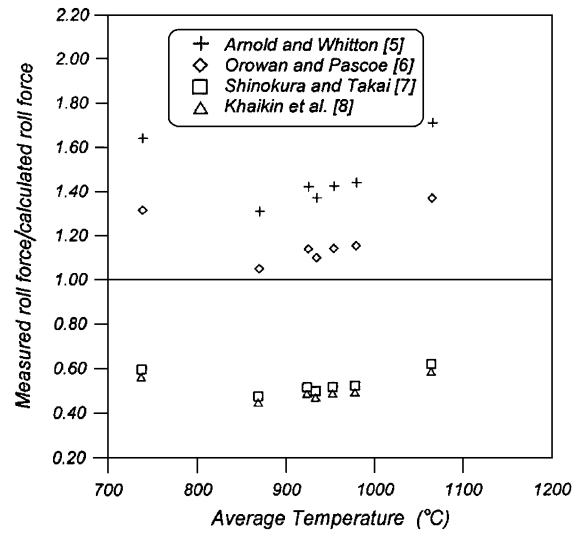


Fig. 8. Comparison of the predicted and measured roll forces for square–diamond pass rolling (reprinted from Said et al. [4], Fig. 6, p. 152).

forces. This is not surprising because Said et al. [4] also showed that the roll forces calculated from Shinokura and Takai's method over-valued the measured force by about 40% in a square–diamond pass rolling experiment (Fig. 8). One of the reasons for this difference might be that Shinokura and Takai tried to include so many rolling pass sequences (oval–round, square–diamond, box–oval, diamond–diamond, square–oval and oval–oval) within a single regression curve. It should be noted that Shinokura and Takai also used Shida's equation [10] to calculate the deformation resistance of material. Another reason might be the method for computing the strain at a pass. Shinokura and Takai [5] simply used the area strain model, taking the natural logarithm of the reduction ratio of the cross-sectional area of the workpiece through the rolling passes. The strain calculated from this area strain model might be quite undervalued.

6. Concluding remarks

The focus of this study has been on the development of a semi-analytical method for the computation of roll force in a round–oval (or oval–round) pass sequence. The concept of a weak plane-strain deformation condition has been introduced to formulate the equation for the roll force. The projected contact area was modeled by a polynomial function and the deformation resistance of the material was characterized by Shida's equation [10].

Hot rod rolling experiment was conducted to examine the predictive capabilities of the proposed semi-analytical method to the temperature variation of the material. Good agreement between the measured roll forces and the predicted forces was found and also for the temperature variations. It is concluded that the proposed semi-analytical method not only has simplicity and accuracy for practical usage, but also saves a large amount of computational time compared with the finite element method.

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